

1. Basics of Engineering Mechanics

Engineering Mechanics deals with the behavior of physical bodies when subjected to forces.

Branches:

- Statics – Study of bodies at rest.
- Dynamics – Study of bodies in motion.

Important Terms:

- Force: Push or pull acting on a body.
- Mass: Quantity of matter.
- Weight: Force due to gravity ($W = mg$).
- Particle and Rigid Body concepts.

2. Force Systems

Types of Forces:

- Collinear forces
- Concurrent forces
- Parallel forces
- Coplanar forces

Resultant Force:

Single force replacing multiple forces.

Equilibrium Conditions:

$$\Sigma F_x = 0$$

$$\Sigma F_y = 0$$

$$\Sigma M = 0$$

3. Free Body Diagram (FBD)

FBD is a diagram showing all forces acting on a body.

Steps:

1. Isolate the object.
2. Show all applied forces.
3. Show reactions.
4. Resolve forces into components.

4. Friction

Types:

- Static Friction
- Kinetic Friction

Laws of Friction:

- Friction opposes motion.
- Depends on normal reaction.

Coefficient of Friction:

$$\mu = F / N$$

Angle of Friction and Angle of Repose concepts.

5. Work, Energy and Power

Work = Force $\times$ Distance

Energy = Capacity to do work.

Types:

- Kinetic Energy = $\frac{1}{2} mv^2$
- Potential Energy = mgh

Power = Work / Time

Unit: Watt

6. Centroid and Moment of Inertia

Centroid is the geometric center of a body.

Moment of Inertia:

Resistance to rotation.

For Rectangle:

$$I = bd^3 / 12$$

Parallel Axis Theorem.

7. Simple Machines

Types:

- Lever
- Pulley
- Screw Jack
- Wheel and Axle

Mechanical Advantage = Load / Effort

Efficiency = MA / VR

8. Trusses

Truss is a structure of connected members.

Assumptions:

- Members connected by pin joints.
- Loads applied at joints.

Methods:

- Method of joints
- Method of sections

9. Kinematics

Study of motion without forces.

Terms:

- Velocity
- Acceleration
- Displacement

Equations of Motion:

$$v = u + at$$

$$s = ut + \frac{1}{2} at^2$$

$$v^2 = u^2 + 2as$$

10. Important Tips for Exams

- Draw neat diagrams.
- Write formulas clearly.
- Use units.
- Practice numericals.
- Learn derivations of basic formulas.